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Are stablecoins truly diversifiers, hedges, or safe havens against traditional cryptocurrencies as their name suggests?

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ABSTRACT

We study the diversifier, hedge, and safe haven properties of stablecoins against traditional cryptocurrencies. Using the DCC-GARCH model and dummy variable regression, we examine the risk-dispersion abilities of USD-pegged and gold-pegged stablecoins against traditional cryptocurrencies, and also compare their risk-dispersion abilities with their underlying assets. The empirical results show that (i) stablecoins can serve as safe havens in specific situations, although most act merely as an effective diversifier in normal market conditions; (ii) gold-pegged stablecoins perform worse as safe havens than USD-pegged ones, but both of them perform better than their corresponding underlying assets, and (iii) the safe haven property of stablecoins changes across market conditions. The above results are robust when using time-varying copula models. We further evaluate the risk management applications of stablecoins by analyzing mixed cryptocurrency-stablecoin portfolios, finding evidence of USD-pegged stablecoins performing better than gold-pegged stablecoins in extreme risk reductions. Our work provides important insights into diversification for cryptocurrency investors.

1. Introduction

1.1. Background

Cryptocurrency (e.g., Bitcoin) is designed as a decentralized peer-to-peer payment system which allows online payments to flow directly from one party to another without going through a financial institution. Owing to these attractive characteristics, cryptocurrency has received much attention in recent years. A recent survey reported that more than a third of American investors are willing to invest in or hold Bitcoin.[1] However, due to the drastic fluctuation of cryptocurrency prices, investors with weak risk tolerance hesitate to add cryptocurrencies to their portfolios. Fig. 1 depicts the daily closing prices of Bitcoin (BTC), Litecoin (LTC), and Ripple (XRP) from August 5, 2013 to March 20, 2019, obtained from coinmarketcap.com. For instance, the value of Bitcoin has skyrocketed to over ten million times in nine years from its introduction in 2009 to its peak price of nearly $20,000 in December 2017. However, during the next six months, Bitcoin depreciated by nearly 70%, and the total market capitalization of cryptocurrencies shrank from a peak value of $840 billion in January 2018 to $261.7 billion in September 2018. Thus, the existence of giant bubbles in the cryptocurrency markets has been investigated (Cheah and Fry, 2015; Fry and Cheah, 2016; Corbet et al., 2018).

The high-volatility character of cryptocurrencies makes it difficult for investors to gain stable returns or maintain value. Thus, there is a pressing need for proper investment tools to hedge against traditional cryptocurrencies. In this context, stablecoins were

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[1] See https://grayscale.co/insights/bitcoin-investor-study-2019.

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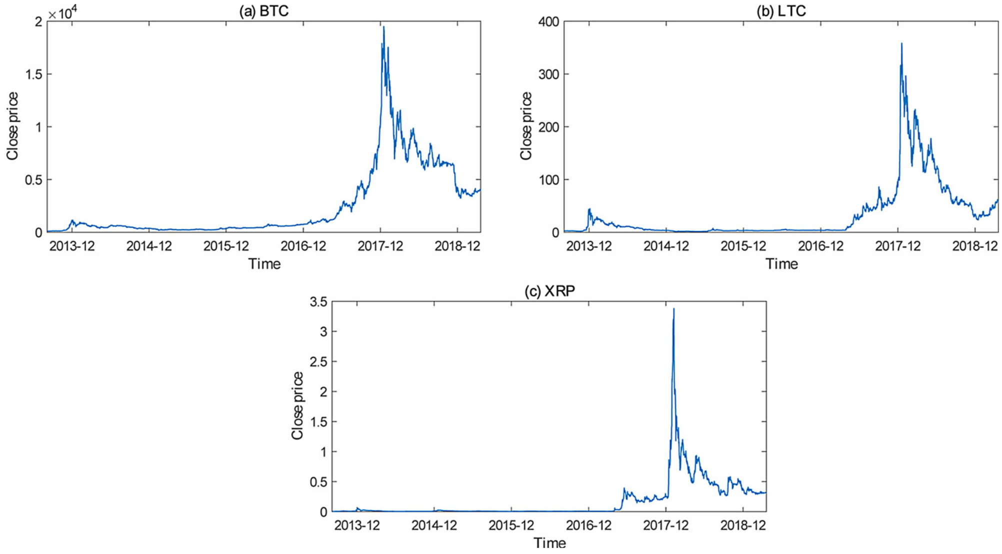

Fig. 1. Daily closing prices of BTC, LTC, and XRP from August 5, 2013 to March 20, 2019. Source: coinmarketcap.com.

introduced as an alternative to traditional cryptocurrencies. Because of the difference in the conception of their technology, stablecoins differ greatly from traditional cryptocurrencies, both in design and in investors' perceptions of them. A useful currency has three key functions: as a store of value, a unit of account, and a medium of exchange. Stability is a key premise for these three functions because: (i) investors are reluctant to store their wealth in a volatile asset whose value constantly changes, (ii) pricing goods becomes difficult with a fluctuating currency, and (iii) people hope for fair payment for goods and services without changes in price during the payment process. Therefore, volatile assets like traditional cryptocurrencies cannot function as an effective store of value, a good unit of account, or a medium of exchange. Stablecoins are created to address these issues of traditional cryptocurrencies. As their name implies, stablecoins are designed to be a price-stable cryptocurrency. Further, they are different from traditional cryptocurrencies in terms of investors' perceptions. Owing to their pegging mechanism, stablecoins bridge fiat currencies with traditional cryptocurrencies. Unlike traditional cryptocurrencies, stablecoins could maintain value and hedge risk for other volatile assets, catering to the various needs of different investors.

Specifically, a stablecoin is pegged either to a fiat currency (e.g., USD and CNY), or to a commodity (e.g., precious metals such as gold and silver). Since there is a network of decentralized vaults and commodity holders, stablecoins are always decentralized, which makes them more attractive to cryptocurrency users. The market demand for low-volatility digital currency tools has accelerated the rapid development of stablecoins. Currently, there are 23 types of stablecoins in circulation, of which many (e.g., Tether, DGD, HGT, and XAUR) are fiat-collateralized, some (e.g., Dai, BitUSD, and BitCNY) are crypto-collateralized, and a few (e.g., Havven and NuBits) are non-collateralized.[2] According to Mita et al. (2019), (i) broadly speaking, stablecoins are a stabilization mechanism achieved by controlling the proportional relationship of exchange rate between traditional cryptocurrencies and fiat currencies, and (ii) pegging is an effective way to reduce asset volatility. Sidorenko (2020) also notes that the cryptocurrency market trend is moving in the direction of transferring funds to some representative low-volatility digital assets, which affirms the ability of stablecoins to save or exchange market assets.

Since stablecoins are designed as low-volatility cryptocurrencies against traditional high-volatility ones to avoid the risk of market declines, two questions naturally present themselves: what potential role would stablecoins play in the fierce fluctuations in the prices of traditional cryptocurrencies? Would they perform the roles of their anchors (e.g., gold and USD) as diversifiers, hedges, or safe havens against traditional cryptocurrencies?

Many previous works (Baur and Lucey, 2010; Baur and McDermott, 2010; Ranaldo and Söderlind, 2010; Bekiros et al., 2017; Wen and Cheng, 2018) have proved the hedge or safe haven functions of gold or USD relative to traditional assets through different models. Based on this, and to answer the first question, our paper studies whether low-volatility stablecoins have the same functions as traditional assets. To highlight the low-volatility effect, we focus on the diversifier, hedge, and safe haven properties of stablecoins against traditional cryptocurrencies, for instance, the most outstanding stablecoin, Tether, against the most representative cryptocurrency, Bitcoin. On the one hand, when stablecoins are strong hedging assets for traditional cryptocurrencies, this means that they are not related to traditional cryptocurrencies under normal market conditions. On the other hand, when stablecoins are strong safe haven assets for traditional cryptocurrencies, this means that stablecoins are irrelevant to traditional cryptocurrencies under extreme market conditions, which also implies that stablecoins can be applied to build investment portfolios superior to traditional

[2] Note that on June 18, 2019, Facebook announced that Libra, a stablecoin fully backed by a reserve of assets, will launch (along with 27 other partners) in 2020. The Libra project has caused wide public concerns; for instance, the US Congress wanted Facebook to halt the development of the Libra project due to regulatory hurdles.

cryptocurrencies. Further, to answer the second question, we compare the potential functions of the low-volatility stablecoins with traditional hedging or safe-haven assets (i.e., gold or USD) against traditional cryptocurrencies.

It is worth noting that Chohan (2019) argues that whether stablecoins are truly stable is still an unresolved question. This is because, in addition to technical or political factors, the credibility of pegged currencies and resources is also uncertain, which will affect the demand of stablecoins for market investors to a certain extent. Therefore, we consider this in our research on the stable function of stablecoins.

Specifically, to examine whether stablecoins are diversifiers, hedges, or safe havens against traditional cryptocurrencies, we follow Baur and Lucey (2010) and Baur and McDermott (2010) in using the dynamic conditional correlation (DCC)-GARCH model and the dummy variable regression. Like in Reboredo (2013b), we also evaluate their risk management functions for investors with analysis of portfolios including a stablecoin and a traditional cryptocurrency. Finally, we employ time-varying copulas to test the robustness of our findings.

1.2. Literature review

Our work is related to the literature on whether gold and cryptocurrencies (especially Bitcoin) are diversifiers, hedges, or safe havens against traditional assets (e.g., stocks, USD, bonds and commodities). As a traditional class of assets, gold has often been considered a safe haven. Much attention has been focused on whether gold is a diversifier, a hedge, or a safe haven in the literature. Baur and Lucey (2010) first discuss the testable definition of a diversifier, a hedge, and a safe haven, and pioneer the test of this hypothesis by building a regression equation of gold returns on extreme returns of stocks and bonds. According to their study, gold can be regarded as a hedge against stocks on average and a safe haven in extreme stock market conditions, though the safe haven property is short-lived. Following Baur and Lucey (2010), a large body of research has tested the diversification properties of gold from different perspectives (Baur and McDermott, 2010; Gürgün and Ünalmış, 2014). Moreover, other approaches (e.g., copula models, quantile regressions, wavelet analysis, and smooth transition) are also used to detect the diversifier, hedge, and safe haven properties of gold (see, e.g., Ciner et al., 2013; Reboredo, 2013a,b; Beckmann et al., 2015; Bekiros et al., 2017; Dar and Maitra, 2017; Śmiech and Papież, 2017; Wen and Cheng, 2018).

With rapid changes in market capitalization and increasing media attention to cryptocurrencies in recent years, there has been a growing research interest in the economics and finance of this new kind of asset (Corbet et al., 2019; Wang et al., 2019a), particularly Bitcoin. Moreover, several studies focus on their market efficiency (Vidal-Tomás and Iba¨nez, 2018; Zhang et al., 2018; Zargar and Kumar, 2019), information spillover effects among cryptocurrencies (Yi et al., 2018; Beneki et al., 2019; Katsiampa, 2019; Sifat et al., 2019; Omane-Adjepong and Alagidede, 2019), and investor attention to Bitcoin (Dastgir et al., 2019; Shen et al., 2019).

Since Bitcoin has shown great resilience in times of turmoil, an increasing number of papers discuss the potential roles of Bitcoin against traditional assets (i.e., a safe haven, a hedging asset, or a diversifier) and draw different conclusions. Many studies (Feng et al., 2018; Stensás et al., 2019; Urquhart and Zhang, 2019) find Bitcoin to be a great diversifier for stocks, commodities, and certain currencies, while others (Bouri et al., 2017a; Stensás et al., 2019; Wang et al., 2019b) observe that in particular situations, cryptocurrencies such as Bitcoin can act as safe haven or hedging tools for commodity indices, major stock indices, and currencies. However, Kliber et al. (2019) argue that Bitcoin plays different roles across countries based on their analysis of five countries. Further, they find that the transaction currency also matters; for instance, Bitcoin can be regarded as a weak hedge when traded in USD. In addition, a few works (see, e.g., Selmi et al., 2018; Bouoiyour et al., 2018) compare the hedging and safe haven properties of Bitcoin with gold, finding that both serve as safe haven assets in particular times (e.g., political and economic turmoil).

In contrast, some studies question the hedge and safe-haven properties of Bitcoin. For example, Bouri et al. (2017b) use a DCC-GARCH model to examine the hedge and safe haven functions of Bitcoin for major world stock indices. They find that the two properties of Bitcoin vary across horizons and in the majority of cases, Bitcoin acts as only a poor hedge. Shahzad et al. (2019) use a bivariate cross-quantilogram approach to study the safe haven property of Bitcoin, gold, and commodity against stocks and find that each can only act as a weak safe haven asset in some cases. Baur et al. (2018) investigate the attributes of Bitcoin and show that it is mainly used as a speculative investment rather than as a currency and medium of exchange. Smales (2019) analyzes Bitcoin's correlations with other assets and its price discovery, volatility, and liquidity characteristics and concludes that it is not worth considering Bitcoin as a safe haven.

Since Bitcoin lost more than half of its market share in the cryptocurrency market,[3] much attention has been paid to the capabilities of other cryptocurrencies against traditional assets. Bouri et al. (2019) find that other leading cryptocurrencies including Ethereum and Litecoin can serve as effective diversifiers as well as hedges, especially against Asian Pacific and Japanese equities, suggesting that investors should consider other choices besides Bitcoin alone. Wang et al. (2019c) investigate the hedge and safe haven properties of 973 types of cryptocurrencies against 30 international indices, and observe that cryptocurrency is a safe haven but not a hedge in general.

The above discussion indicates that the characteristics of cryptocurrencies could be controversial, and the diversification capabilities of cryptocurrencies as represented by Bitcoin are contentious. Nevertheless, it is certain that cryptocurrencies are extremely volatile and subject to speculative behavior from investors, which means that before being treated as a diversification tool, they should be diversified first. For investors who hold cryptocurrencies for capital gains rather than hedging other physical assets, a critical issue could be how to diversify, hedge, and eliminate extreme losses from cryptocurrencies.

[3] According to Bouri et al. (2019), the market share of Bitcoin shrank from around 90% in early 2017 to 48% in July 2018.

As a type of cryptocurrency, stablecoins are designed to address the wild price swings of traditional cryptocurrencies and bridge fiat currencies with traditional cryptocurrencies (Mita et al., 2019; Sidorenko, 2020). Stablecoins offer users a way to enjoy the benefits of virtual currencies while also partially reduce their volatility risks. Recently, an increasing number of investors have been attracted by the low-volatility character of stablecoins. High market demand for low-volatility digital assets has already led to a remarkable jump in the stablecoin economy (Sidorenko, 2020). According to a report by blockchain.com, the total market share of stablecoins in the cryptocurrency market increased from 1.5% in September 2018 to 2.7% in February 2019 (Micky News, 2019). Compared with traditional cryptocurrencies, stablecoins are preferred by market participants or investors who (i) are not willing to be exposed to additional currency risk when transacting in cryptocurrencies; (ii) need a tool to minimize risk in the turnover of cryptocurrencies without withdrawing funds back to a bank fiat account; (iii) solely seek a store of value on a censorship-resistant ledger to avoid the local banking system or collapsing economies.

Since traditional cryptocurrencies are extremely volatile and stablecoins are designed to address this problem, traditional cryptocurrency investors have the motivation to add stablecoins into their portfolios to diversify, hedge, and eliminate extreme losses. However, little attention has been paid to stablecoins in the literature, especially their potential against traditional cryptocurrencies. Our work aims to fill this gap by investigating the diversifier, hedge, and safe haven properties of stablecoins against traditional cryptocurrencies.

1.3. Main contributions

Much research has examined the role of gold as a hedge or a safe haven against different traditional assets, and a growing number of studies focus on the relations between cryptocurrencies (especially Bitcoin) and traditional assets; however, research on the potential roles of stablecoins against traditional cryptocurrencies is scarce. Our work builds on the framework of Baur and Lucey (2010) and Baur and McDermott (2010), but the targets and methods adopted in our study are different. We choose three USD-pegged stablecoins (Tether, BitUSD, and NuBits), three gold-pegged stablecoins (DGD, HGT, and XAUR), and three traditional cryptocurrencies (Bitcoin, Litecoin, and Ripple). Our research contributes to the existing literature by examining the diversifier, hedge, and safe haven roles of stablecoins against traditional cryptocurrencies. In addition, we examine whether stablecoins help to reduce loss for cryptocurrency investors in extreme market situations by considering a cryptocurrency-stablecoin portfolio. In greater detail, our contributions are as follows.

First, we study the dependence structure between stablecoins and traditional cryptocurrencies using the DCC-GARCH model of Engle (2002), from which we obtain dynamic conditional correlations of the pairs of assets under study. We employ ARMA-GARCH models to fit marginal distributions of returns of stablecoins and traditional cryptocurrencies, and use the DCC model to estimate dynamic correlations between stablecoins and traditional cryptocurrencies.

Second, to accurately analyze the dynamic correlations between stablecoins and cryptocurrencies in different stages, we broaden the research using multiple structural change models from Bai and Perron (1998, 2003) to divide the time horizon of traditional cryptocurrencies (BTC, LTC, XRP) into three stages. We examine two aspects: (i) which stablecoin perform better as the role of diversifier, hedge, or safe haven, and (ii) whether the function of stablecoin will change across market stages.

Third, since the interdependence between stablecoins and traditional cryptocurrencies is useful for portfolio investors who seek to reduce risk, we evaluate the value-at-risk (VaR) and excepted shortfall (ES) reduction of portfolios that consist of traditional cryptocurrencies and stablecoins with different weights. This evaluation helps determine which combinations reduce downside risk to investors. Combining with the DCC-GARCH model and downside risk evaluations, we find that (i) USD-pegged stablecoins perform better as hedges and safe havens against traditional cryptocurrencies than gold-pegged ones and (ii) Tether is a good asset for reducing losses caused by the decline of traditional cryptocurrencies.

Finally, as a supplement, we use time-varying copulas to calculate the dynamic dependence between stablecoins and traditional cryptocurrencies. The results we obtain from the estimation of copulas are consistent with our findings under the DCC framework.

1.4. Organization

The paper is structured as follows. Section 2 outlines the methodologies. Section 3 describes the data and Section 4 presents empirical results of the DCC-GARCH model and dummy variable regression, which includes both the full-period and subperiod analyses. Section 5 shows the risk management analysis of different-weighted portfolios. Section 6 reports the robustness test based on time-varying copulas. Section 7 summarizes our findings and conclusions.

2. Methodology

2.1. The DCC-GARCH model

We use the dynamic conditional correlation (DCC)-GARCH model of Engle (2002) to capture dynamic relationships across time series. In this study, a bivariate DCC-GARCH model is employed for pairs of return series to avoid biased estimates of parameters. The estimation of the bivariate DCC-GARCH model is carried out in two steps. In the first step, an ARMA-GARCH model is estimated to fit the marginal distributions of returns. In the second, a time-varying correlation matrix is computed using the standardized residuals obtained from the first step estimation.

We first use different ARMA-GARCH models or ARMA-GJR-GARCH models with different distributions to fit marginal distributions of returns for different stablecoins and traditional cryptocurrencies. In some cases, we establish a seasonal differential processing on the return series before the ARMA-GARCH estimation to eliminate the disturbance of seasonal factors. The ARMA(p, q)-GARCH(1,1) model is specified as[4]

$$r_{i,t} = \mu_i + \sum_{j=1}^{p} \varphi_{i,j} r_{i,t-j} + \sum_{j=1}^{q} \phi_{i,j} \varepsilon_{i,t-j} + \varepsilon_{i,t} \tag{1}$$

$$\sigma_{i,t}^2 = c_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2 \tag{2}$$

$$\eta_{i,t} = \frac{\varepsilon_{i,t}}{\sigma_{i,t}} \tag{3}$$

In Eq. (1) (i.e., mean equation), $r_{i,t}$ denotes the returns of stablecoins or traditional cryptocurrencies i, μ_i is the constant term, $\varphi_{i,j}$ and $\phi_{i,j}$ are autoregressive and moving average parameters, and $\varepsilon_{i,t}$ is the residual series. In Eq. (2) (i.e., variance equation), $\sigma_{i,t}^2$ is the conditional variance, c_i is the constant, α_i is the parameter that captures the short-run persistence or the ARCH effect and β_i is the parameter that captures the long-run persistence of volatility or the GARCH effect. In Eq. (3), $\eta_{i,t}$ is the standardized residual series following a specific i.i.d. distribution.[5]

To model the asymmetry volatility of stablecoins and traditional cryptocurrencies, we consider the GJR-GARCH(1,1) model,[6] which is defined as

$$\sigma_{i,t}^2 = c_i + \alpha_i \varepsilon_{i,t-1}^2 + \gamma_i \varepsilon_{i,t-1}^2 I_{i,t-1} + \beta_i \sigma_{i,t-1}^2 \tag{4}$$

where $I_{i,t-1}$ is a dummy variable, which is used to set a threshold to distinguish the impact of positive and negative shocks on conditional volatility.

As in Engle (2002), the DCC-GARCH(1,1) equation is specified for the conditional covariance matrix $\boldsymbol{Q}_t$ with element $q_{\mathrm{ij},t}$, which is a positive-definite matrix:

$$\boldsymbol{Q}_t = (1 - \theta_1 - \theta_2)\bar{\boldsymbol{Q}} + \theta_1 \boldsymbol{Q}_{t-1} + \theta_2 \boldsymbol{\eta}_{t-1} \boldsymbol{\eta}_{t-1}' \tag{5}$$

where θ_1 and θ_2 are non-negative parameters that satisfy $\theta_1 + \theta_2 < 1$ and capture the influences of previous DCCs and previous shocks on the current DCC respectively; $\bar{\boldsymbol{Q}}$ is the unconditional covariance matrix of $\boldsymbol{\eta}_t$; and $\boldsymbol{\eta}_t = (\eta_{1t}, \eta_{2t})'$ represents a marginally standardized innovation vector, that is, the vector of standardized residuals obtained from the estimation of (GJR)-GARCH(1,1) process.

The DCC matrix is calculated by:

$$\boldsymbol{\rho}_t = \boldsymbol{J}_t \boldsymbol{Q}_t \boldsymbol{J}_t \tag{6}$$

where $\boldsymbol{J}_t = \mathrm{diag}\{q_{11,t}^{-1/2}, q_{22,t}^{-1/2}\}$ is a normalized matrix to ensure that $\boldsymbol{\rho}_t$ is a correlation matrix, and $q_{\mathrm{ii},t}$ is the (i, i)th element of matrix $\boldsymbol{Q}_t$.

The pairwise DCC between stablecoin i and traditional cryptocurrency j is given by:

$$\rho_{\mathrm{ij},t} = \frac{q_{\mathrm{ij},t}}{\sqrt{q_{\mathrm{ii},t}}\sqrt{q_{\mathrm{jj},t}}} \tag{7}$$

2.2. Dummy variable regression

The role of stablecoins as diversifiers, hedges, or safe havens against traditional cryptocurrencies are examined using the dummy regression model as in Baur and McDermott (2010), Ratner and Chiu (2013) and Bouri et al. (2017b). Dynamic conditional correlations $\rho_{\mathrm{ij},t}$ between stablecoin i and traditional cryptocurrency j are extracted from the bivariate DCC-GARCH model into separate time series and then regressed on dummy variables indicating extreme downward movements in the lower 10th, 5th, or 1st percentile of the return distribution of traditional cryptocurrency j. The dummy variable regression model is specified as:

$$\rho_{\mathrm{ij},t} = m_0 + m_1 D(r_{\mathrm{crypto}}\, q_{10}) + m_2 D(r_{\mathrm{crypto}}\, q_5) + m_3 D(r_{\mathrm{crypto}}\, q_1) + v_t \tag{8}$$

where $\rho_{\mathrm{ij},t}$ represents the pairwise conditional correlations between stablecoin i and traditional cryptocurrency j, v_t is the error term,

[4] About the specification of the ARMA component, we mainly use the AMRA(1,1), ARMA(2,1) or ARMA(2,2) model to achieve a simple and good fitting effect and eliminate the autocorrelation in returns. In the Online Appendix, Tables A1–A6 report the detailed estimation results of ARMA-GARCH(1,1) or ARMA-GJR-GARCH(1,1) models for return series of traditional cryptocurrencies and stablecoins during different periods.

[5] In our work, four distributions are considered to fit the standardized residual series of returns for traditional cryptocurrencies and stablecoins: Gaussian distribution, student's t-distribution, the generalized error distribution (GED) and skewed student's t-distribution. For the returns of the traditional cryptocurrencies and stablecoins during the full-sample period, most of them always follow a Gaussian distribution, except that (i) the return series of NuBits perfectly obeys a skewed student's t-distribution and (ii) that of HGT and XAUR follows a GED. But for the subperiod sample, the best fitted distributions for different returns are diverse and respectively obey the above four distributions. However, most of the returns are subject to a Gaussian distribution or a student's t-distribution. The best fitted marginal distributions of returns for the full-period sample and the subperiod sample can be found in Tables A1–A6 in the Online Appendix.

[6] Note that according to McAleer (2014) and Caporin and Costola (2019), the GJR-GARCH model is asymmetric but fails to capture the leverage effect.

Table 1
Introduction to USD-pegged stablecoins.

	Tether	BitUSD	NuBits
Tickers	USDT, EURT	BITUSD	USNBT
Launch date	2014	2014	2014
Market cap	\$2,803,538,394	\$4,468,672	\$587,517
Top-level category	Asset-collateralized	Asset-collateralized	Algorithmic
Reference peg	USD	USD	USD
Country location	Distributed	US	Sweden
Platform	Omni protocol	Unknown	Unknown

$r_{\text{crypto}}q_{10}$, $r_{\text{crypto}}q_5$, and $r_{\text{crypto}}q_1$ are 10%, 5%, and 1% quantiles of returns for traditional cryptocurrency j, and $D(\cdot)$ is a dummy variable for obtaining the extreme price movement of traditional cryptocurrency j. Taking $D(r_{\text{crypto}}q_{10})$ as an example, it takes 1 if the return of traditional cryptocurrency j is smaller than the 10% quantile, otherwise it takes 0.

Following Baur and McDermott (2010), Ratner and Chiu (2013) and Bouri et al. (2017b), we determine whether stablecoin i is a diversifier, a hedge or a safe haven against traditional cryptocurrency j by analyzing the estimated values and statistical significance of m_i $(i = 0, 1, 2, 3)$ in Eq. (8). In summary, there are five cases: (i) if m_0 is significantly positive, stablecoin i (or the other assets under study) is a diversifier against traditional cryptocurrency j; (ii) if m_0 is zero, stablecoin i is a weak hedge against traditional cryptocurrency j; (iii) if m_0 is significantly negative, stablecoin i is a strong hedge; (iv) if m_1, m_2 and m_3 are not significantly different from zero, stablecoin i is a weak safe haven against traditional cryptocurrency j; and (v) if m_1, m_2 and m_3 are negative, stablecoin i is a strong safe haven.

3. Data and primary analysis

Considering both the liquidity and market capitalization, we select three representative USD-pegged stablecoins including Tether, BitUSD, and NuBits, and three typical gold-pegged stablecoins including DGD, HGT, and XAUR. Tables 1 and 2 present the detailed information on these six stablecoins. In what follows, we show a brief introduction to them.

USD-pegged stablecoins can be exchanged with USD usually at a 1:1 ratio. Tether was initially designated to be worth \$1.00, and equal amounts of dollars must be deposited to support its issuance. However, on March 14, 2019 it was announced that only \$0.74 cash was backing each Tether, which meant that the value of Tether had shrunk by 26%. BitUSD, which is linked with the blockchain of BitShares, is backed by at least twice the amount calculated based on BitShares' price. The essence of the BitShares system is the Market Anchoring Mechanism, which aims to make market prices of BitUSD ultimately consistent with those of the USD under the excess of BitShares. NuBits is the world's first stablecoin designed to keep its value at \$1.00. Thus, the supply of NuBits is dynamically adjusted in response to the changes in its demand. NuBits experienced ups and downs in demand, and remained infamously underperforming for 3 months in 2016 and thereafter grew 1.5 times by the end of 2017.

Gold-pegged stablecoins are backed by gold and can be exchanged for physical gold. DGD is constructed by DigixDAO (a decentralized autonomous institution) and is disposed in the blockchain by DigixGlobal. DGD is issued as an ICO (Initial Coin Offerings), which provides holders with the corresponding voting rights according to the amount of DGD they hold. HGT, which is a HelloGold project, corresponds to the rights and interests of HelloGold platform's trading of handling and custody fee, because HelloGold's business model includes a gold financial service platform based on gold backed tokens (GBT) and equity certificates based on HGT. Its operation is similar to that of DGD. XAUR, which is in the POS blockchain and switches to Ethereum, is created for the exchange of a unit physical gold.

In addition to USD-pegged and gold-pegged stablecoins, we use three traditional cryptocurrencies, including Bitcoin (BTC), Litecoin (LTC) and Ripple (XRP), to determine the potential roles of stablecoins against traditional cryptocurrencies. According to the cryptocurrency market capitalization as of March 20, 2019, BTC, XRP, and LTC are the first, third, and fifth largest cryptocurrencies, respectively. We do not consider the second and fourth largest cryptocurrencies, ETH and Bitcoin Cash, because their starting trading dates are later than the start date of our investigated sample period. The data on stablecoins and traditional cryptocurrencies are from

Table 2
Introduction to gold-pegged stablecoins.

	HelloGold	DigixDAO	Xaurum
Tickers	GOLDX, HGT	DGD	XAUR
Launch date	2017	Early 2014	2016
Market cap	\$206,641	\$76,042,442	\$2,824,259
Top-level category	Asset-collateralized	Asset-collateralized	Asset-collateralized
Reference peg	1 g of gold	1 g of gold	1 g of gold
Country location	Malaysia	Singapore	Slovenia
Platform	Ethereum	Ethereum	Ethereum

Table 3
Descriptive statistics of returns for two samples.

	Obs.	Mean	Std. dev.	Min.	Max.	Skewness	Kurtosis	Jarque-Bera	ADF
Panel A: USD-pegged sample									
Tether	1438	0.000	0.014	−0.047	0.500	29.125	1011.361	61,126,098***	−99.939***
BitUSD	1438	0.000	0.057	−0.772	0.734	−0.436	83.911	392,291***	−28.057***
NuBits	1438	−0.002	0.058	−0.358	0.924	3.450	61.411	207,278***	−17.706***
BTC	1438	0.002	0.038	−0.208	0.225	−0.298	8.472	1815***	−37.942***
LTC	1438	0.002	0.059	−0.514	0.510	0.799	17.601	12,927***	−38.008***
XRP	1438	0.002	0.072	−0.616	1.027	3.086	45.073	108,341***	−23.437***
USD	1042	0.000	0.000	−0.021	0.025	−0.098	5.073	188***	−32.184***
Panel B: Gold-pegged sample									
DGD	524	−0.003	0.077	−0.460	0.580	0.404	12.148	1841***	−23.313***
HGT	524	−0.005	0.201	−0.918	1.249	0.657	7.833	547***	−20.444***
XAUR	524	−0.003	0.070	−0.296	0.544	0.477	10.645	1296***	−28.967***
BTC	524	−0.001	0.044	−0.185	0.225	−0.003	6.049	202***	−22.278***
LTC	524	0.000	0.027	−0.092	0.169	1.212	9.988	1194***	−22.465***
XRP	524	0.000	0.075	−0.353	0.607	1.945	18.147	5339***	−13.605***
Gold	360	0.000	0.006	−0.020	0.024	0.015	3.992	15***	−20.492***

Notes: Jarque-Bera statistic tests for the null hypothesis of normal distribution. ADF is a statistic of the augmented Dickey–Fuller test for a unit root.
*** Denotes the rejection of null hypothesis at the 1% significance level.

coinmarketcap.com. For comparison, we also take into account the underlying assets of USD-pegged and gold-pegged stablecoins, USD and gold. We use the US Dollar Index (USDX) and the Gold Index of London (AUUSDO) to represent USD and gold.

Because launch dates for the two types of stablecoins are different (see Tables 1 and 2), the sample periods that we collect for USD-pegged and gold-pegged stablecoins are not identical. USD-pegged stablecoins were launched earlier than gold-pegged ones, and their sample period is from March 6, 2015 to March 20, 2019, or 1438 observations. The starting date is determined by the day when Tethers returns changed from zeros to floating values. The sample period for gold-pegged stablecoins starts on October 13, 2017 and ends on March 20, 2019, or 524 observations. We collect the daily closing prices of the six stablecoins in the corresponding sample period. In each sample period, we also collect the daily closing prices of the three traditional cryptocurrencies (i.e., BTC, LTC, and XRP) and the corresponding underlying-asset (i.e., USD or gold). For simplicity, we designate the sample from March 6, 2015 to March 20, 2019 including three USD-pegged stablecoins, three traditional cryptocurrencies, and USD as the USD-pegged sample, and the other sample from October 13, 2017 to March 20, 2019, including three gold-pegged stablecoins, three traditional cryptocurrencies, and gold as the gold-pegged sample.

Table 3 shows the descriptive statistics of returns for the six stablecoins and three traditional cryptocurrencies as well as for their underlying assets, USD and Gold. We find that NuBits and HGT show the highest volatility among USD-pegged and gold-pegged stablecoins, respectively. Among traditional cryptocurrencies, XRP has the highest return and standard deviation in both samples (see Panels A and B). Contrary to our expectations, the returns of stablecoins, especially gold-pegged stablecoins, also fluctuate conspicuously. The returns of cryptocurrencies are quite extreme; for example, the maximum return of HGT reaches 125% while its minimum return is low to 92%. As discussed, stablecoins were created to reduce the risk of excessive fluctuations in traditional cryptocurrencies. Thus, when the market is under stress, the prices of stablecoins should remain unchanged or only fluctuate slightly. However, the huge fluctuations of HGT show that stablecoins cannot fully realize stable functions in extreme market conditions. The high kurtosis of the virtual assets under study indicates a great number of tail returns, and the results of non-zero skewness reflect asymmetric distributions of returns. The Jarque-Bera test results imply that none of the returns follow Gaussian distribution, which is in accordance with results drawn from skewness and kurtosis. USD and Gold, as traditional safe haven assets, present some features that are distinct from those of cryptocurrencies. Their returns are much less volatile than those of both stablecoins and traditional cryptocurrencies. The distributions of their returns are close to the Gaussian distribution, with a skewness near 0 and a kurtosis near 3. The ADF unit root tests show that all returns are stationary, indicating that they can be modeled on ARMA-GARCH or ARMA-GJR-GARCH.

4. Empirical results

4.1. Dynamic conditional correlations

4.1.1. Full period analysis

Fig. 2 presents the full-period dynamic conditional correlations (DCCs) between three prevalent USD-pegged stablecoins (i.e., Tether, BitUSD, and NuBits) and three representative traditional cryptocurrencies (i.e., BTC, LTC and XRP). Fig. 2(c), (f), and (i) depicts DCCs between NuBits and three traditional cryptocurrencies. We find that the DCCs are almost all positive, suggesting that in most times NuBits is positive correlated to traditional cryptocurrencies. In particular, the DCCs between NuBits and LTC and between NuBits and XRP show a similar trend. The DCCs between BitUSD and traditional cryptocurrencies are also positive as the whole (see Fig. 2(b), (e), and (h)), but their DCCs are highly volatile. Fig. 2(a), (d), and (g) shows that the dynamic correlation results for Tether

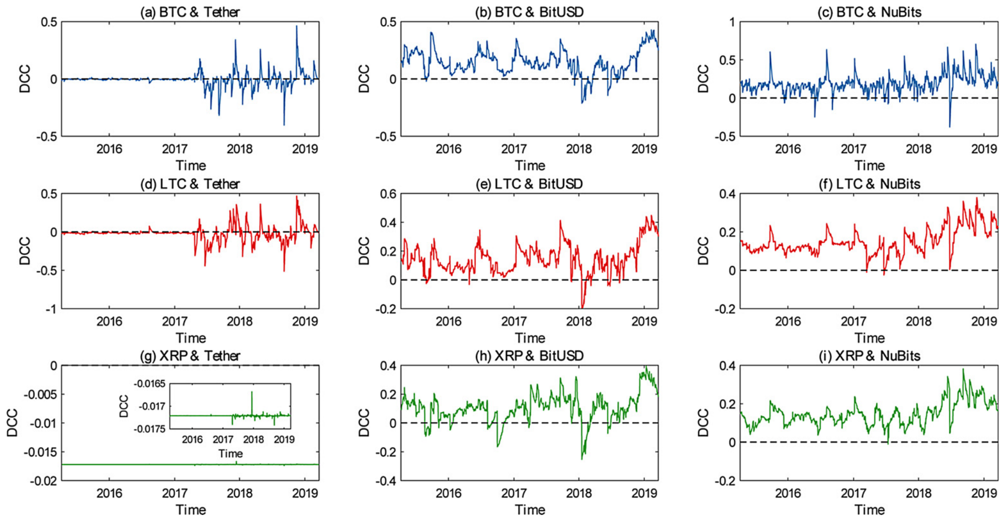

Fig. 2. DCCs between USD-pegged stablecoins and traditional cryptocurrencies from full sample analysis. Notes: Because the change of DCCs between XRP and Tether is not obvious in the range [−0.02, 0], we insert a secondary subgraph into panel (g) to magnify the fluctuation of DCCs.

are noteworthy. The DCCs between Tether and three traditional cryptocurrencies are nearly stable and equal to 0 before April 2017, and have been fluctuating thereafter. The main reason of these pronounced structural changes in the DCC processes may be the following unique properties of Tether. Namely, the closing prices of Tether are nearly fixed at $1.00 before April 2017, thus its returns are nearly 0 for a relatively long time, partly reflecting the stability of Tether. We observe that the DCCs between Tether and traditional cryptocurrencies are not always positive. Particularly, the dynamic correlations between Tether and XRP are negative throughout the sample period.

In summary, according to the full-period DCCs between USD-pegged stablecoins and traditional cryptocurrencies in Fig. 2, we find that (i) BitUSD and NuBits are positive-related to traditional cryptocurrencies in most times and (ii) Tether seems to have the best risk-dispersion effects.

Fig. 3 shows the full-period DCCs between three gold-pegged stablecoins (i.e., HGT, DGD, and XAUR) and three traditional cryptocurrencies (i.e., BTC, LTC, and XRP). Unlike USD-pegged stablecoins, the DCCs between gold-pegged stablecoins and three traditional cryptocurrencies are almost always positive. This result suggests that the hedge effect of the three gold-pegged stablecoins against traditional cryptocurrencies is quite limited, especially for DGD, because its dynamic correlations with the three traditional cryptocurrencies exceed 0.5 at most times.

In short, based on the DCCs between gold-pegged stablecoins and traditional cryptocurrencies, we find little hedging effect of

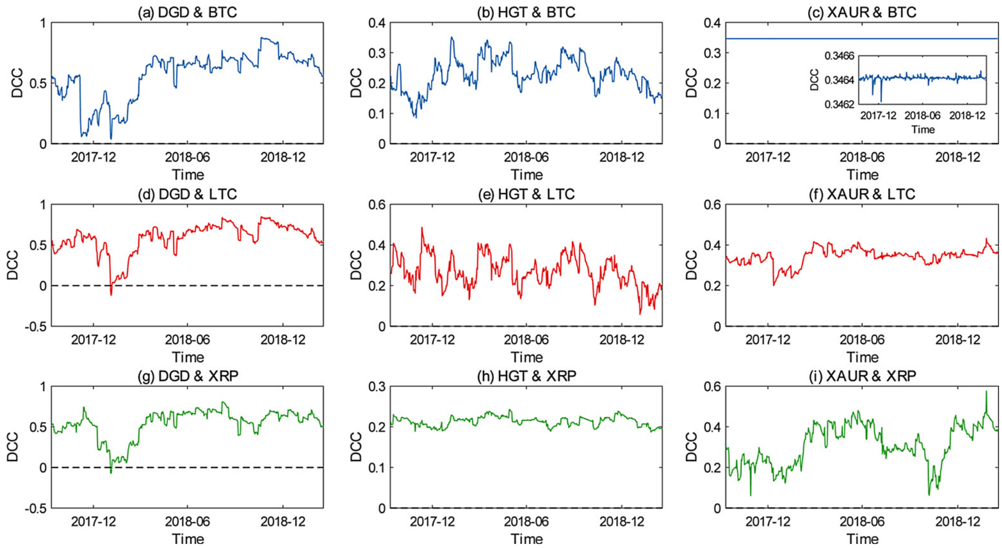

Fig. 3. DCCs between Gold-pegged stablecoins and traditional cryptocurrencies from full sample analysis. Notes for this figure: Because the change of DCCs between BTC and XAUR is not obvious in the range [0, 0.4], we insert a secondary subgraph into panel (c) to magnify the fluctuation of DCCs.

Table 4
Breakpoints in returns of three traditional cryptocurrencies (i.e., BTC, LTC, and XRP) from March 6, 2015 to March 20, 2019.

Break	BTC	LTC	XRP
1	25/03/2017	28/03/2017	18/03/2017
2	17/12/2017	19/12/2017	08/01/2018

gold-pegged stablecoins against traditional cryptocurrencies, which suggests that gold-pegged stablecoins may not be a perfect class of assets for cryptocurrency investors to build a hedge portfolio.

4.1.2. Subperiod analysis

The second half of 2017 witnessed the boom of cryptocurrencies including BTC, LTC, and XRP; for example, over several months, the price of one BTC rose from $2500 to $19,000. However, this was not a lasting trend: cryptocurrencies experienced dramatic declines in 2018, and the value of BTC declined by more than 50%. In this context, to further examine whether the diversifier, hedge, or safe haven roles of stablecoins are affected by market environments, we check the robustness over the full-period sample using multiple structural change models from Bai and Perron (1998, 2003) to detect breakpoints in the returns of the three traditional cryptocurrencies. Specifically, we use the sequential $L+1$ breaks vs. L method from Bai and Perron (1998, 2003) to detect breakpoints in the returns of BTC, LTC, and XRP, and divide the full period into subperiods before and after the breakpoints.

Table 4 shows breakpoints in the returns of three traditional cryptocurrencies over the full period from March 6, 2015 to March 20, 2019 (corresponding to the USD-pegged sample period). Note that here we only consider the USD-pegged sample period because of a short gold-pegged sample period. We detect two breakpoints in the returns of each traditional cryptocurrency over the entire period, meaning that the entire period is divided into three subperiods in terms of each traditional cryptocurrency. Based on the breakpoints of each traditional cryptocurrency, we divide the USD-pegged sample period into three subperiods, which correspond to the boom, the depression, and the recovery, respectively.

Fig. 4 shows the DCCs between USD-pegged stablecoins and BTC in three subperiods. During subperiod 1, DCCs between (i) Tether and BTC fluctuate between −0.2 and 0.2, and most of the correlations are negative, (ii) BitUSD and BTC are all positive, reaching the peak value over 0.4, and (iii) NuBits and BTC float sharply in the range $[-0.5, 1]$. During subperiod 2, Tether and BTC are negatively correlated in most cases, while DCCs between another two stablecoins (i.e., BitUSD and NuBits) and BTC stabilize at 0.117 and 0.046, respectively. During subperiod 3, we find that (i) DCCs between Tether and BTC fluctuate around zero and the maximum value is close to 0.5, (ii) DCCs between BitUSD and BTC are mostly negative during the first half subperiod, but become positive and show an increasing trend during the other half, and (iii) NuBits is positively correlated with BTC (except for isolated cases).

From the above various results obtained from the DCC analysis in Fig. 4, we find that DCCs between USD-pegged stablecoins and BTC are time-varying and stablecoins have different DCC patterns with traditional cryptocurrencies. In particular, two of the three USD-pegged stablecoins (i.e., BitUSD and NuBits) present statically positive correlations with BTC during subperiod 2, implying that they can be influenced by the extreme downside of BTC. It is a reminder that regardless of their special issuing mechanism,

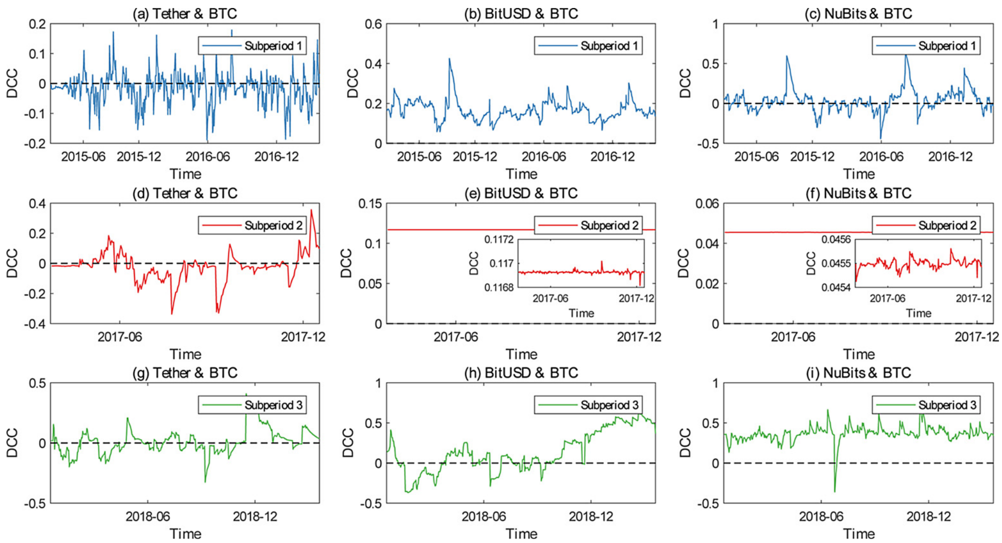

Fig. 4. DCCs between USD-pegged stablecoins and BTC from subperiod analysis. Notes: Because the change of DCCs between BTC and BitUSD is not obvious in the range [0, 0.15], we insert a secondary subgraph into panel (e) to magnify the fluctuation of DCCs. Similarly, we insert a secondary subgraph into panel (f) to magnify the fluctuation of DCCs between BTC and NuBits.

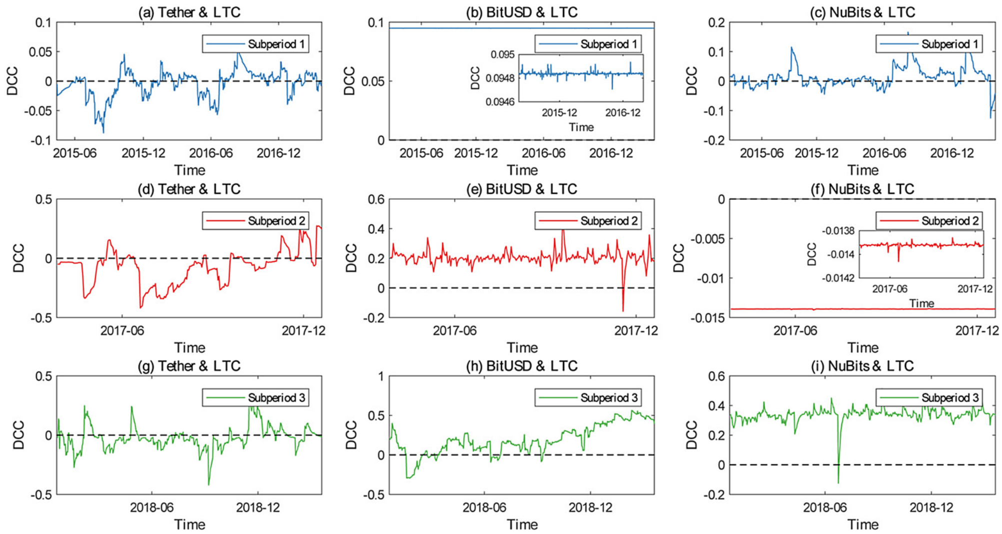

Fig. 5. DCCs between USD-pegged stablecoins and LTC from subperiod analysis. Notes: Because the changes of DCCs between LTC and BitUSD is not obvious in the range [0, 0.1], we insert a secondary subgraph into panel (b) to magnify the fluctuation of DCCs. Similarly, we insert a secondary subgraph into panel (f) to magnify the fluctuation of DCCs between LTC and NuBits.

stablecoins are still one class of the virtual assets. The relations between stablecoins and traditional cryptocurrencies are quite like cross-sector relations in stock markets. The slump of a "giant" stock may lead to a fall in the others, including those from different sectors. However, we should note the special case of Tether, because it is negatively correlated with BTC during all three subperiods in general.

Fig. 5 depicts the DCCs between USD-pegged stablecoins and LTC in each subperiod. Overall, the results are similar to that shown in Fig. 4, except that during subperiod 2, NuBits maintains a negative correlation with LTC at −0.014. Also, the DCCs between BitUSD and LTC are close to 0.1 during the first subperiod and remain prominently positive during the next two stages.

Fig. 6 exhibits the DCCs between USD-pegged stablecoins and XRP in different subperiods. In contrast to the results obtained from Figs. 4 and 5, the dynamic correlations between Tether and XRP stabilize at 0.046 during subperiod 2, which reflects some particular traits of XRP and suggests that it is difficult for investors to avoid the risk of a fall in XRP by investing in USD-pegged stablecoins.

4.2. Dummy variable regression

In this section, we use the dummy variable regression to check whether stablecoins are diversifiers, hedges or safe havens for

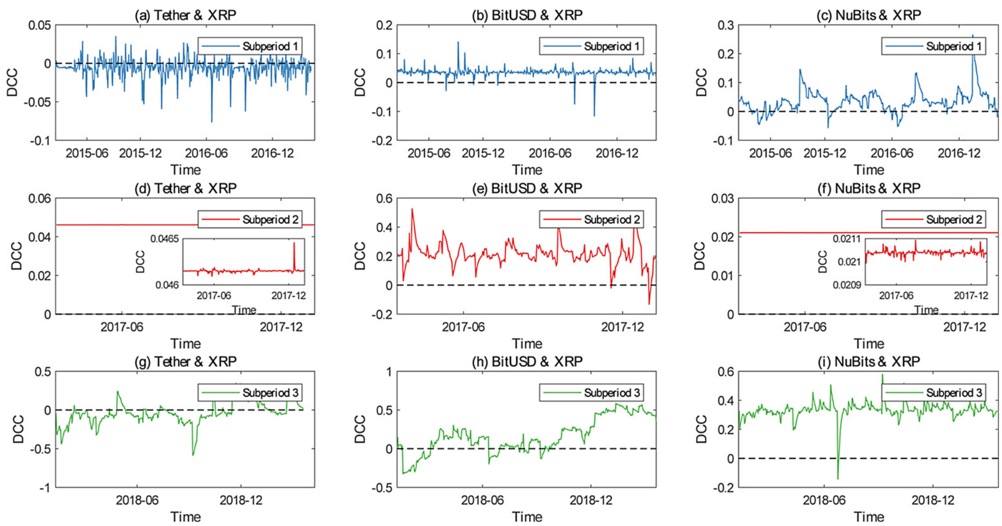

Fig. 6. DCCs between USD-pegged stablecoins and XRP from subperiod analysis. Notes: Because the change of DCCs between XRP and Tether is not obvious in the range [0, 0.06], we insert a secondary subgraph into panel (d) to magnify the fluctuation of DCCs. Similarly, we insert a secondary subgraph into panel (f) to magnify the fluctuation of DCCs between XRP and NuBits.

Table 5
Estimation results on the diversifier, hedge, and safe haven properties of stablecoins against traditional cryptocurrencies (i.e., BTC, LTC, and XRP) in the entire period based on DCC-GARCH models.

		Hedge (m_0)	10% quantile (m_1)	5% quantile (m_2)	1% quantile (m_3)
Panel A: Potential roles of stablecoins against BTC					
USD-pegged	Tether	−0.004**	−0.009	0.019*	0.007
	BitUSD	0.152***	−0.060***	0.026	−0.010
	NuBits	0.192***	0.016	−0.001	0.023
	USD	−0.010***	0.000	0.000**	0.000
Gold-pegged	DGD	0.580***	−0.026	−0.068	−0.105**
	XAUR	0.346***	0.000	0.000*	0.000
	HGD	0.230***	−0.005	−0.015	0.001
	Gold	−0.050***	0.000	0.000	0.000
Panel B: Potential roles of stablecoins against LTC					
USD-pegged	Tether	−0.019***	−0.012	0.034**	−0.004
	BitUSD	0.149***	−0.031**	0.023	−0.013
	NuBits	0.151***	0.013	−0.004	−0.003
	USD	0.042***	0.000	0.000**	0.000
Gold-pegged	DGD	0.597***	−0.080*	0.023	−0.080
	XAUR	0.345***	−0.014	0.001	−0.017
	HGD	0.271***	−0.002	−0.025	−0.004
	Gold	0.020***	0.000	0.000	0.000
Panel C: Potential roles of stablecoins against XRP					
USD-pegged	Tether	−0.017***	0.000	0.000	0.000
	BitUSD	0.107***	−0.024**	0.027	−0.038
	NuBits	0.147***	−0.001	0.005	−0.028
	USD	−0.040***	0.000**	0.000	0.000**
Gold-pegged	DGD	0.553***	−0.117***	0.032	−0.170*
	XAUR	0.314***	−0.050*	0.030	−0.130**
	HGD	0.214***	−0.002	−0.002	−0.007
	Gold	0.076***	−0.013	0.047	−0.009

* Denote the rejection of null hypothesis at the 10% significance level.
** Denote the rejection of null hypothesis at the 5% significance level.
*** Denote the rejection of null hypothesis at the 1% significance level.

traditional cryptocurrencies. We examine the potential roles of USD-pegged and gold-pegged stablecoins over the entire sample period. Further, we consider the potential roles of USD-pegged stablecoins in subperiods corresponding to different market conditions.

4.2.1. Full-period analysis

In Table 5, we report estimated coefficients of Eq. (8) for examining the diversifier, hedge, and safe haven properties of stablecoins in the entire period. We first focus on the estimated coefficient m_0 to analyze the diversifier or hedge properties of stablecoins against traditional cryptocurrencies. The estimated m_0 of Tether are all negative at the 5% significance level, which are −0.004 for BTC, −0.019 for LTC, and −0.017 for XRP, suggesting the hedge role of Tether against traditional cryptocurrencies. The estimates of BitUSD are 0.152 for BTC, 0.149 for LTC, and 0.107 for XRP at the 1% significance level, which indicate that BitUSD can be a diversifier for them. Like BitUSD, the estimated coefficients of NuBits are all positive at the 1% significance level, also implying the diversifier property of NuBits against traditional cryptocurrencies. The diversifier property also holds for the three gold-pegged stablecoins, because their estimated coefficients m_0 are positive at the 1% significance level. To summarize, all stablecoins are diversifiers for the three traditional cryptocurrencies, except Tether, which is a strong hedge asset for them.

We then focus on the estimated coefficients m_1, m_2, and m_3 (corresponding to the 10%, 5% and 1% quantiles, respectively) to explore the safe haven property of stablecoins against traditional cryptocurrencies. For extreme downward situations, most estimated coefficients m_1, m_2, and m_3 are insignificant. For example, none of the estimates of NuBits or HGT are significant for BTC, LTC, or XRP at any of the three quantiles. However, a few estimates are significant at certain quantile levels. For USD-pegged stablecoins, the estimated m_2 of Tether for BTC are 0.019 at the 10% significance level and 0.034 for LTC at the 5% significance level, indicating that Tether can be a safe haven asset for BTC and LTC. The estimated m_1 of BitUSD for BTC, LTC, and XRP are all significantly negative. Specifically, the estimated coefficients m_1 of BitUSD are −0.060 for BTC at the 1% significance level, and −0.031 and −0.024 for LTC and XRP at the 5% significance level, respectively, indicating that BitUSD can act as a safe haven asset for traditional cryptocurrencies (especially for BTC) in the 10% downward situations.

For gold-pegged stablecoins, we have the following findings: (i) the estimated m_2 of XAUR for BTC is zero at the 10% significance level, suggesting that XAUR can be a weak safe haven for BTC; (ii) the estimated coefficient m_1 of DGD for BTC is −0.105 at the 5% significance level, which implies DGD can be considered a strong safe haven for BTC; (iii) the estimated m_1 and m_3 of XAUR and DGD

Table 6
Estimation of diversifier, hedge, and safe haven properties of stablecoins against traditional cryptocurrencies (i.e., BTC, LTC, and XRP) from the subperiod analysis.

	BTC				LTC				XRP			
	m_0	m_1	m_2	m_3	m_0	m_1	m_2	m_3	m_0	m_1	m_2	m_3
Panel A: Subperiod 1												
Tether	−0.016***	0.000	−0.003	0.014	−0.004***	0.007	−0.007	−0.009	−0.006***	0.001	−0.003	0.003
BitUSD	0.166***	0.005	0.001	−0.022	0.095***	0.000	0.000	0.000	0.035***	0.000	0.003	−0.002
NuBits	0.039***	−0.010	0.001	0.073	0.014***	−0.009	0.016*	0.013	0.035***	0.002	−0.003	−0.011
Panel B: Subperiod 2												
Tether	−0.018**	−0.009	−0.045	0.040	−0.072***	0.031	−0.043	−0.045	0.046***	0.000	0.000	0.000
BitUSD	0.012***	0.000	0.000	0.000	0.205***	0.002	−0.003	−0.030	0.213***	0.008	−0.001	−0.036***
NuBits	0.046***	0.000	0.000	0.000	−0.014***	0.000	0.000	0.000	0.021***	0.000**	0.000***	0.000
Panel C: Subperiod 3												
Tether	0.013*	0.009	0.005	0.005	−0.025***	0.017	0.022	0.021	−0.036***	0.003	0.001	−0.095
BitUSD	0.144***	−0.116	0.096	−0.118*	0.195***	−0.056	0.042	−0.059	0.171***	−0.071	0.015	0.003
NuBits	0.367***	0.009	−0.017	0.011	0.338***	0.043	0.001	−0.002	0.331***	0.011	−0.022	−0.006

* Denote the rejection of null hypothesis at the 10% significance level.
** Denote the rejection of null hypothesis at the 5% significance level.
*** Denote the rejection of null hypothesis at the 1% significance level.

for XRP are all significantly negative, implying that they can act as a strong safe haven for XRP; and (iv) all estimated m_1, m_2, and m_3 of gold-pegged stablecoins for LTC are insignificant, suggesting that gold-pegged stablecoins are not safe havens for LTC.

Notably, there are significant differences among the three USD-pegged stablecoins with regard to the diversifier, hedge, and safe haven properties. Tether can be regarded as a hedge on average, but as a diversifier in extreme conditions. Although BitUSD co-moves with traditional cryptocurrencies, it is a strong safe haven at certain quantiles. NuBits is neither a hedge nor a safe haven against traditional cryptocurrencies. One convincing explanation for these differences may be that different USD-pegged stablecoins actually have very different blockchain algorithms, though they are backed by the same asset (i.e., USD). Namely, Tether is fiat-collateralized, BitUSD is crypto-collateralized, and NuBits is basically algorithmic and non-collateralized. Different algorithms, levels of transparency, and market positions may result in various indications.

The reason stablecoins sometimes act as safe havens for traditional cryptocurrencies is worth pondering. A possible explanation is that when extreme downward market conditions occur, investors transfer their money from traditional cryptocurrencies to some stablecoins and this leads to the appreciation of these stablecoins. Another potential interpretation could be that when negative shocks appear, investors transfer their money from cryptocurrencies to traditional safe haven assets (e.g., USD and gold) to reduce risks, which leads to the price increase of USD and gold, and then accordingly causes the appreciation of corresponding stablecoins.

4.2.2. Subperiod analysis

Table 6 shows the estimated coefficients of Eq. (8) during three subperiods. We find that the diversifier, hedge, and safe haven properties of stablecoins (i.e., USD-pegged stablecoins) are not static and change across different subperiods. Specifically, during subperiod 1, the estimated coefficients m_0 of Tether are negative for all traditional cryptocurrencies at the 1% significance level, which are −0.016 for BTC, −0.004 for LTC, and −0.006 for XRP. The estimates of BitUSD and NuBits are all positive at the 1% significance level. These results confirm the findings in the full-sample analysis, including the hedge role of Tether and the diversifier property of BitUSD and NuBits against traditional cryptocurrencies. However, the safe haven property of USD-pegged stablecoins are disappointing during subperiod 1, because none of them can be regarded as a weak or strong safe haven against extreme downside movements of the three traditional cryptocurrencies.

Since subperiod 1 witnessed the boom of cryptocurrencies and a major increase in blockchain investments, returns on cryptocurrencies during this stage were quite fruitful. In contrast to the sharp rise of traditional cryptocurrencies, the prices of stablecoins are much less volatile. Therefore, the correlations between the two kinds of assets are smaller than those in the full sample period. On the other hand, cryptocurrencies barely show any movement or dramatic decline in this subperiod, and stablecoins cannot exert their safe haven function against extreme downside risks.

During subperiod 2, the estimated coefficients m_0 of Tether are −0.018 for BTC, −0.072 for LTC and 0.046 for XRP, showing that Tether acts as a strong hedge for BTC and LTC. BitUSD shows no hedge effectiveness for three traditional cryptocurrencies in this subperiod because all estimates of m_0 are significantly positive. NuBits is an effective diversifier for BTC and XRP, and it acts as a strong hedge for LTC because its estimated coefficient m_0 is −0.014 at the 1% significance level.

When it comes to extreme downward situations during subperiod 2, the estimated coefficients m_1, m_2, and m_3 of Tether are insignificant for all traditional cryptocurrencies, indicating that Tether has no safe haven function during this crisis period. BitUSD appears to be a strong safe haven for XRP with a negative estimated m_3 (i.e., −0.036) at the 1% significance level. NuBits can be regarded as a weak safe haven asset for XRP in the 10% and 5% extreme downward conditions because the estimates of m_1 and m_2 equal to zero at the 5% significance level.

Because traditional cryptocurrencies experienced dramatic declines during subperiod 2, the results of this stage reveal the diversifier, hedge, and safe haven properties of USD-pegged stablecoins under an extreme downward market condition. During this crisis period, none of the three stablecoins can be regarded as a strong safe haven for BTC or LTC. Since BTC is the most influential cryptocurrency, its price behavior has a great impact on the other digital currency assets. Thus, the sharp decline of BTC influences the behavior of stablecoins and eliminates their effectiveness in diversifying extreme downside risk. Nevertheless, BitUSD and NuBits have a safe haven function for XRP. This finding implies that during times of crisis, XRP investors could invest in USD-pegged stablecoins to avoid risk. In terms of international influence and liquidity, USD is the most powerful currency in the world currency system, and it is widely acknowledged as a safe haven asset (see, e.g., Wen and Cheng, 2018). Therefore, USD-pegged stablecoins are always regarded as virtual asset with stable intrinsic value. Investors may have the motivation to turn to USD-pegged stablecoins when XRP meets adverse movements, and the increasing demand would push up the prices of stablecoins.

During subperiod 3, the cryptocurrency market gradually rebounded and became less fluctuant than the first two stages. The estimated coefficients m_0 of Tether are 0.013 for BTC, −0.025 for LTC, and −0.036 for XRP, which indicate that Tether can be regarded as a strong hedge against XRP and LTC, and it is only a diversifier for BTC. Regarding BitUSD, the estimates of m_0 are 0.144 for BTC, 0.195 for LTC, and 0.171 for XRP. These three estimates are significant at the 1% level, suggesting that BitUSD is no more than an effective diversifier for traditional cryptocurrencies on average. Like BitUSD, the estimated coefficients m_0 of NuBits are all positive at the 1% significance level, which implies that NuBits is also a diversifier for traditional cryptocurrencies on average. As for extreme downward situations, most estimates of m_1, m_2, and m_3 are insignificant. The only exception is the estimated coefficient m_3 of BitUSD for BTC, which is −0.118 at the 10% significance level. This indicates the strong safe haven property of BitUSD for BTC at the 1% extreme downward quantile.

The subperiod results are consistent with those of the full sample analysis, but they also indicate that the market condition matters to the diversifier, hedge, and safe haven properties of USD-pegged stablecoins. Our main findings are as follows: (i) Tether is a strong hedge in most conditions except for BTC in subperiod 3 and for XRP in subperiod 2. There is no significant estimate of Tether as a safe haven in any subperiods, indicating that Tether has no safe haven property in extreme downward conditions. (ii) BitUSD is a diversifier in all conditions but shows a strong safe haven property for XRP at the 1% quantile in the crisis period and for BTC at the same quantile in subperiod 3. (iii) In most situations, NuBits is a diversifier except for LTC in subperiod 2, and it can be regarded as a weak safe haven for XRP at the 10% and 5% quantiles in the crisis period.

4.3. Evidence for USD and gold

Next, we build regression models to compare the diversifier, hedge, and safe haven properties of stablecoins with those of their underlying assets, USD and gold. Table 5 shows the estimated results of USD and gold from the dummy variable regression (i.e., Eq. (8)).

The estimated m_0 of USD are −0.010 for BTC, 0.042 for LTC, and −0.040 for XRP at the 1% significance level, indicating that USD can be a strong hedge for BTC and XRP, but can be no more than an effective diversifier against LTC. The estimated m_0 of gold is −0.050 for BTC, 0.020 for LTC, and 0.076 for XRP, which implies that gold is a strong hedge for BTC and is a diversifier against LTC and XRP.

For extreme negative returns on traditional cryptocurrencies, the estimated coefficients of USD are zero for BTC and LTC in the 5% extreme market condition (i.e., m_2), and for XRP at the 10% and 1% quantiles (i.e., m_1 and m_3) at the 5% significance level. However, all estimates m_1, m_2, and m_3 of Gold are not significant at any levels. There are several interesting findings from the above results. (i) USD performs a little better than gold in the hedge function, because USD can be regarded as a strong hedge for BTC and LTC, while gold is a strong hedge only for BTC. (ii) The two traditional assets have limited safe haven effects for traditional cryptocurrencies. USD can only act as a weak safe haven for BTC, LTC, and XRP for certain quantiles and gold is no more than a diversifier in all situations. In some way, this means that returns of USD are not impacted by extreme downward risks of cryptocurrencies. (iii) USD serves as a better hedge than BitUSD and NuBits, but a worse one than Tether, while its safe haven properties are not as strong as that of BitUSD. This shows that USD has closer connections with BTC, LTC, and XRP on average than Tether, and has closer connections with cryptocurrencies during the crisis period than BitUSD. (iv) Gold does act as a better hedge tool for traditional cryptocurrencies than stablecoins it backs. However, we find that gold is not a safe haven for traditional cryptocurrencies.

On the whole, stablecoins act better than their underlying assets as safe havens. The better safe haven property of stablecoins may be because stablecoins were introduced due to the huge price fluctuations of traditional cryptocurrencies, which serve as a medium of exchange between the worldwide digital currency and the fiat currency. Thus, it is more convenient for investors to transfer their money from traditional cryptocurrencies to stablecoins than to USD or gold due to lower transaction fees and the basic blockchain technology they share.

5. Implications for risk management

To understand whether stablecoins function as a risk-dispersion tool in extreme market events and reduce the losses for investors, we construct different asset combinations and apply the extreme value theory (EVT) approach to the following portfolio analysis. Under the EVT framework, we use the peaks-over-threshold (POT) method with the generalized pareto distribution (GPD) to calculate the value-at-risk (VaR) and excepted shortfall (ES) of the portfolios under study.

First, we consider a portfolio, called portfolio 1, obtained by minimizing the risk of a cryptocurrency-stablecoin portfolio without reducing the expected return. Following Kroner and Ng (1998) and Reboredo (2013b), the optimal weight of the stablecoin in portfolio 1 at time t is specified as

$$\omega_t^S = \frac{h_t^C - h_t^{S-C}}{h_t^S - 2h_t^{S-C} + h_t^C} \tag{9}$$

where ω_t^S is between 0 and 1. Specifically, if $\omega_t^S > 1, \omega_t^S = 1$, and if $\omega_t^S < 0, \omega_t^S = 0$. h_t^S, h_t^C and h_t^{S-C} are the conditional volatility of a stablecoin, the conditional volatility of a traditional cryptocurrency and the conditional covariance between them at time t, respectively. ω_t^S is the weight of a stablecoin and the weight of a traditional cryptocurrency is equal to $1 - \omega_t^S$.

Besides portfolio 1, we build a passively managed portfolio 2, which gives the same weight to a stablecoin and a traditional cryptocurrency, and the weight at time t is specified as

$$\omega_t^S = \frac{1}{2} \tag{10}$$

Given a portfolio composed of a traditional cryptocurrency and a stablecoin, its return can be expressed as

$$r_t = \log(\omega_t^S e^{r_t^S} + (1 - \omega_t^S)e^{r_t^C}) \tag{11}$$

where r_t^S, r_t^C, and r_t are returns of a stablecoin, a traditional cryptocurrency and their portfolio, respectively.

The next step is to calculate VaR and ES of the three traditional cryptocurrencies as well as portfolios established before. We employ the POT approach from Davison and Smith (1990) to measure extreme risks for the traditional cryptocurrencies and portfolios under study. Practically, the measurement comprises two steps: first, a GPD is used to fit the excess distribution of the returns of each asset (traditional cryptocurrency and portfolio) with the POT method to extract extremes; second, two extreme risk measures VaR and ES are estimated as extreme quantiles of the GPD.

According to the POT approach, the excess distribution of returns r_t based on the threshold η can be approximated by the GPD as:

$$G_{\xi,\psi(\eta)}(x) = 1 - \left[1 + \frac{\xi x}{\psi(\eta)}\right]^{-1/\xi} \tag{12}$$

where x represents the exceedance over the threshold η, ξ is the shape parameter, and $\psi(\eta) > 0$ is the scale parameter. The threshold is selected to ensure that a given percentage of extremes lies above it.

The estimated parameters in Eq. (12) are applied to compute the two risk measurements VaR and ES as:

$$\mathrm{VaR}_q = \eta - \frac{\psi(\eta)}{\xi}\left\{1 - \left[\frac{T}{N_\eta}(1-q)\right]^{-\xi}\right\} \tag{13}$$

and

$$\mathrm{ES}_q = \frac{\mathrm{VaR}_q}{1-\xi} + \frac{\psi(\eta) - \xi\eta}{1-\xi} \tag{14}$$

where VaR_q represents the qth quantile of returns r_t, ES_q represents the expected loss under the condition that $r_t > \mathrm{VaR}_q$, T represents the sample size of returns, and N_η is the number of exceedances over the threshold η.

The final step is to compare the VaR and ES of a cryptocurrency-stablecoin portfolio, including a traditional cryptocurrency and a stablecoin with those of only a traditional cryptocurrency in extreme markets, so that we can know whether the stablecoin really plays a safe haven role against the traditional cryptocurrency. To make the calculation results easier to compare, we introduce the VaR or ES reduction measure, which is the difference between the VaR or ES of a traditional cryptocurrency and that of a

Table 7
Downside risk evaluation for portfolios against BTC.

	Tether	BitUSD	NuBits	USD	DGD	HGT	XAUR	Gold
Portfolio 1								
VaR reduction (95%)	0.053	0.019	0.021	0.059	−0.007	−0.009	0.003	0.067
ES reduction (95%)	0.078	0.027	0.022	0.088	−0.013	−0.013	0.004	0.095
VaR reduction (99%)	0.094	0.036	0.021	0.107	−0.016	−0.017	0.004	0.113
ES reduction (99%)	0.114	0.031	0.028	0.134	−0.024	−0.011	0.009	0.138
VaR reduction (99.9%)	0.139	0.022	0.037	0.170	−0.035	−0.001	0.016	0.171
ES reduction (99.9%)	0.149	−0.031	0.049	0.188	−0.045	0.010	0.023	0.193
Portfolio 2								
VaR reduction (95%)	0.032	0.016	0.007	0.034	−0.017	−0.071	0.058	0.041
ES reduction (95%)	0.049	0.019	0.004	0.051	−0.024	−0.100	0.002	0.057
VaR reduction (99%)	0.059	0.027	0.000	0.061	−0.023	−0.117	0.004	0.066
ES reduction (99%)	0.077	0.004	0.006	0.081	−0.052	−0.143	0.008	0.082
VaR reduction (99.9%)	0.100	−0.025	0.015	0.109	−0.092	−0.177	0.013	0.104
ES reduction (99.9%)	0.118	−0.120	0.028	0.133	−0.168	−0.201	0.017	0.121

Table 8
Downside risk evaluation for portfolios against LTC.

	Tether	BitUSD	NuBits	USD	DGD	HGT	XAUR	Gold
Portfolio 1								
VaR reduction (95%)	0.072	0.031	0.031	0.079	−0.018	−0.015	−0.013	0.030
ES reduction (95%)	0.109	0.049	0.041	0.126	−0.024	−0.017	−0.015	0.044
VaR reduction (99%)	0.127	0.061	0.041	0.149	−0.027	−0.018	−0.017	0.054
ES reduction (99%)	0.183	0.075	0.074	0.218	−0.034	−0.016	−0.017	0.062
VaR reduction (99.9%)	0.257	0.097	0.120	0.309	−0.043	−0.012	−0.018	0.071
ES reduction (99.9%)	0.355	0.084	0.206	0.428	−0.050	−0.009	−0.019	0.074
Portfolio 2								
VaR reduction (95%)	0.042	0.028	0.016	0.045	−0.040	−0.109	−0.037	0.019
ES reduction (95%)	0.065	0.036	0.023	0.072	−0.054	−0.145	−0.077	0.030
VaR reduction (99%)	0.076	0.051	0.021	0.084	−0.057	−0.165	−0.103	0.037
ES reduction (99%)	0.114	0.034	0.056	0.126	−0.097	−0.213	−0.139	0.041
VaR reduction (99.9%)	0.166	0.028	0.105	0.181	−0.151	−0.279	−0.187	0.046
ES reduction (99.9%)	0.242	−0.122	0.197	0.258	−0.252	−0.342	−0.218	0.047

corresponding cryptocurrency-stablecoin portfolio.

Tables 7–9 show the reductions of VaR and ES provided by portfolio 1 and portfolio 2 at the 95%, 99%, and 99.9% confidence levels (corresponding to the 5%, 1% and 0.1% risk levels), which represent the value of loss reduction of USD-pegged stablecoins and gold-pegged ones against three traditional cryptocurrencies. For comparison, we also present results of the underlying-assets (i.e., USD and gold) in Tables 7–9 . We find that reductions of portfolio 1 are much larger than that of portfolio 2, meaning that a positive strategy is better than a passive one that simply divides weights of two assets evenly.

Under the combination of USD-pegged stablecoins, almost all VaR and ES reductions of portfolios are positive, which means that they do help to reduce the investors' losses in extreme downward markets. In addition, Tether provides larger downside risk reductions than NuBits and BitUSD in almost all cases against traditional cryptocurrencies, which implies that Tether is a better safe haven. However, note that the BTC–BitUSD portfolio does not perform well at the 99.9% confidence level with the negative reductions of VaR and ES, indicating limited risk-reduction effects of BitUSD against BTC in the most extremely falling market.

Portfolios containing gold-pegged stablecoins perform much worse than those including USD-pegged ones, which may partly reflect the fact that USD is more stable than gold. Only the LTC–XAUR, XRP–XAUR, and XRP–DGD portfolios show a few positive VaR and ES reductions at different confidence levels. This is consistent with our previous analysis to some extent, of which we detect the safe haven property of DGD and XAUR at specific quantiles. But note that VaR and ES reductions of gold-pegged stablecoins are not as much as those of their underlying asset, gold.

USD-pegged stablecoins provide the highest downside risk reductions for XRP at 99% and 99.9% confidence levels and the lowest for BTC at all confidence levels. Among them, Tether provides the largest reductions that are close to USD and higher than gold. Especially, the VaR and ES reductions of the XRP–Tether portfolio are large as 0.295 and 0.404 at the 99.9% confidence level.

The above results show that no other stablecoins perform better than Tether in acting as a safe haven against traditional cryptocurrencies. Besides, Tether also shows a good value of hedge based on the evaluation of DCCs. Combining both of the conclusions, there is evidence that Tether is absolutely a good asset for risk management on both normal and extreme circumstances.

Table 9
Downside risk evaluation for portfolios against XRP.

	Tether	BitUSD	NuBits	USD	DGD	HGT	XAUR	Gold
Portfolio 1								
VaR reduction (95%)	0.067	0.027	0.029	0.077	0.001	−0.012	0.008	0.091
ES reduction (95%)	0.114	0.057	0.051	0.131	0.012	−0.012	0.022	0.150
VaR reduction (99%)	0.138	0.072	0.058	0.154	0.014	−0.020	0.025	0.184
ES reduction (99%)	0.205	0.111	0.106	0.243	0.049	0.033	0.066	0.253
VaR reduction (99.9%)	0.295	0.163	0.173	0.361	0.098	0.111	0.124	0.344
ES reduction (99.9%)	0.404	0.221	0.271	0.541	0.185	0.229	0.221	0.428
Portfolio 2								
VaR reduction (95%)	0.040	0.023	0.011	0.044	−0.002	−0.062	0.006	0.053
ES reduction (95%)	0.069	0.043	0.029	0.074	0.008	−0.071	0.025	0.087
VaR reduction (99%)	0.084	0.057	0.035	0.087	0.010	−0.082	0.031	0.106
ES reduction (99%)	0.126	0.071	0.078	0.140	0.034	−0.065	0.078	0.147
VaR reduction (99.9%)	0.181	0.093	0.138	0.210	0.067	−0.040	0.144	0.201
ES reduction (99.9%)	0.250	0.079	0.230	0.326	0.128	0.031	0.247	0.253

6. Robustness test

To check the robustness of our results on different dynamic correlation approaches, we use time-varying copulas from Patton (2006) to model the dynamic dependence structure between stablecoins and traditional cryptocurrencies.[7] Based on dynamic dependence coefficients, we then use the dummy variable regression from Eq. (8) to calculate the coefficients that can reflect the diversifier, hedge, and safe haven functions of stablecoins against traditional cryptocurrencies. As we can see from Table 10, the results of Eq. (8) based on time-varying copulas are consistent with those of DCC models. The estimated coefficients m_0 of Tether are −0.012 for BTC and −0.047 for LTC, indicating that Tether behaved as a strong hedge against BTC and LTC. While the other stablecoins can be regarded as no more than an effective diversifier against traditional cryptocurrencies since their estimated m_0 are all positive.

The consistency of the two dynamic correlation approaches also holds for the safe haven property of stablecoins against traditional cryptocurrencies. For USD-pegged stablecoins, the estimated coefficients m_3 of Tether and NuBits are −0.034 and −0.085 for XRP at the 5% significance level, respectively, which implies that Tether and NuBits are both strong safe havens against XRP in the 1% extreme downside market. The estimated m_1 of BitUSD are −0.04 for BTC at the 1% significance level and −0.011 for XRP at the 10% significance level, implying that BitUSD can act as a strong safe haven against BTC and XRP in the 10% downside market. For gold-pegged stablecoins, the estimated coefficients m_1, m_2 and m_3 are all positive under all circumstances of the quantile. In summary, USD-pegged stablecoins can serve as safe havens against traditional cryptocurrencies under specific situations, while gold-pegged stablecoins show no safe haven property under any condition. Namely, the results using time-varying copulas are robust and consistent with our central findings.

7. Conclusions

In this study, we have investigated the diversifier, hedge, and safe haven properties of stablecoins against traditional cryptocurrencies. We first used the ARMA-GARCH models for fitting marginal distributions of each return series under study, and then employed the DCC model to obtain dynamic conditional correlations between stablecoins and traditional cryptocurrencies. We next established dummy variable regressions to examine the potential properties of stablecoins against traditional cryptocurrencies using their dynamic conditional correlations. Both the full-period and subperiod analyses were conducted with these methods. To test the loss-reduction abilities of stablecoins from a risk management perspective, we calculated VaR and ES reductions, defined as the differences between VaR and ES of a traditional cryptocurrency and a cryptocurrency-stablecoin portfolio, respectively. In addition, we also compared the results of the underlying assets (i.e., USD and gold) with those of stablecoins. Our sample data were daily closing prices for three USD-pegged stablecoins (i.e., Tether, BitUSD, and NuBits), three gold-pegged stablecoins (i.e., DGD, HGT, and XAUR), and three traditional cryptocurrencies (i.e., Bitcoin, Litecoin, and Ripple). The sample periods are from March 6, 2015 and October 13, 2017 to March 20, 2019 for analysis of USD-pegged stablecoins and gold-pegged ones, respectively.

We have four major findings as follows. (i) USD-pegged stablecoins have better risk-dispersion abilities for traditional cryptocurrencies than gold-pegged ones according to dynamic conditional correlations. Among USD-pegged stablecoins, Tether shows the best properties of risk diversification, while BitUSD and NuBits are positive-related to traditional cryptocurrencies in most of the time. (ii) Based on the dummy variable regression, we find that Tether can be regarded as a strong hedge for all the three traditional cryptocurrencies from the full-period analysis, while each of other stablecoins is no more than an effective diversifier. However, some of them do show the safe haven property under certain quantile levels. For instance, BitUSD is a strong safe haven against three traditional cryptocurrencies at the 10% quantile and some gold-pegged stablecoins also perform well at certain quantiles. Comparing the results of full-period analysis with the subperiod one, we find that the market condition matters to the risk-dispersion effectiveness of USD-pegged stablecoins. (iii) We find gold acts better as a hedge than the stablecoins it backs, but lacks in the power of safe haven. USD is a better hedge than two of the USD-pegged stablecoins, but it is not as good a safe haven as the stablecoins it backs. (iv) From the risk management perspective, we find that gold-pegged stablecoins hardly have any function in VaR and ES reductions, while their underlying asset gold helps eliminate extreme risks to some extent. Both the USD-pegged stablecoins and USD can be regarded as great tools in reducing extreme losses.

In summary, our work sheds light on the diversifier, hedge, and safe haven properties of USD-pegged and gold-pegged stablecoins against traditional cryptocurrencies from various perspectives, and provides virtual currency investors with comprehensive risk management analysis in normal and extreme downside market conditions. Since the literature on the potential roles (e.g., diversifiers, hedges, or safe haven) of stablecoins is scarce and still evolving, there are three possible directions for future research. The first is to study the hedging ability of stablecoins against traditional assets in different countries. The second is to investigate the safe haven property of stablecoins during political and economic turmoil. The third is to further compare the potential roles of stablecoins with traditional safe haven assets such as gold and USD, in order to explore whether they compete with or complement each other.

Authors' contribution

Gang-Jin Wang: conceptualization, methodology, writing-original draft preparation, writing-reviewing and editing, supervision, and funding acquisition. Xin-yu Ma: data curation, software, investigation, visualization, writing-original draft preparation. Hao-yu

[7] The fitting results of dynamic copula models are reported in the Tables A7–A9 in the Online Appendix.

Table 10
Estimation results on the diversifier, hedge, and safe haven properties of stablecoins against traditional cryptocurrencies (i.e., BTC, LTC, and XRP) in the entire period based on time-varying copulas.

		Hedge (m_0)	10% quantile (m_1)	5% quantile (m_2)	1% quantile (m_3)
Panel A: Potential roles of stablecoins against BTC					
USD-pegged	Tether	−0.012 ***	0.002	0.011	−0.006
	BitUSD	0.181 ***	−0.040 ***	0.027	0.029
	NuBits	0.144 ***	0.024	0.008	−0.039
Gold-pegged	DGD	0.559***	−0.017	−0.033	−0.017
	HGT	0.258***	−0.104	0.003	−0.001
	XAUR	0.415***	0.033	−0.072	0.093
Panel B: Potential roles of stablecoins against LTC					
USD-pegged	Tether	−0.047 ***	−0.012	0.031**	0.000
	BitUSD	0.177 ***	−0.005	0.008	−0.002
	NuBits	0.105 ***	0.046 ***	−0.035	−0.032
Gold-pegged	DGD	0.641 ***	−0.015	0.021	−0.015
	HGT	0.308 ***	−0.008	0.004	−0.012
	XAUR	0.390 ***	0.003	−0.007	0.008
Panel C: Potential roles of stablecoins against XRP					
USD-pegged	Tether	0.000 ***	−0.008	−0.002	−0.034 **
	BitUSD	0.178 ***	−0.011 *	0.011	−0.015
	NuBits	0.141 ***	0.006	0.008	−0.085**
Gold-pegged	DGD	0.581 ***	−0.032	−0.004	−0.096
	HGT	0.247 ***	0.001	−0.004	0.000
	XAUR	0.334 ***	0.003	−0.003	0.014

Notes:
* Denote the rejection of null hypothesis at the 10% significance level.
** Denote the rejection of null hypothesis at the 5% significance level.
*** Denote the rejection of null hypothesis at the 1% significance level.

Wu: data curation, software, investigation, visualization, writing-original draft preparation.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.ribaf.2020.101225.

References

Bai, J., Perron, P., 1998. Estimating and testing linear models with multiple structural changes. Econometrica 66, 47–78. https://doi.org/10.2307/2998540.
Bai, J., Perron, P., 2003. Computation and analysis of multiple structural change models. J. Appl. Econometr. 18, 1–22. https://doi.org/10.1002/jae.659.
Baur, D.G., Hong, K., Lee, A.D., 2018. Bitcoin: medium of exchange or speculative assets? J. Int. Finan. Markets Inst. Money 54, 177–189. https://doi.org/10.1016/j.intfin.2017.12.004.
Baur, D.G., Lucey, B.M., 2010. Is gold a hedge or a safe haven? An analysis of stocks, bonds and gold. Finan. Rev. 45, 217–229. https://doi.org/10.1111/j. 1540-6288.2010.00244.x.
Baur, D.G., McDermott, T.K., 2010. Is gold a safe haven? International evidence. J. Bank. Finance 34, 1886–1898. https://doi.org/10.1016/j.jbankfin.2009.12.008.
Beckmann, J., Berger, T., Czudaj, R., 2015. Does gold act as a hedge or a safe haven for stocks? A smooth transition approach. Econ. Modell. 48, 16–24. https://doi.org/10.1016/j.econmod.2014.10.044.
Bekiros, S., Boubaker, S., Nguyen, D.K., Uddin, G.S., 2017. Black swan events and safe havens: the role of gold in globally integrated emerging markets. J. Int. Money Finance 73, 317–334. https://doi.org/10.1016/j.jimonfin.2017.02.010.
Beneki, C., Koulis, A., Kyriazis, N.A., Papadamou, S., 2019. Investigating volatility transmission and hedging properties between Bitcoin and Ethereum. Res. Int. Business Finance 48, 219–227. https://doi.org/10.1016/j.ribaf.2019.01.001.
Bouoiyour, J., Selmi, R., Wohar, M.E., 2018. Bitcoin: competitor or complement to gold? Econ. Bull. 39, 186–191. http://www.accessecon.com/Pubs/EB/2019/Volume39/EB-19-V39-I1-P19.pdf.
Bouri, E., Jalkh, N., Molnár, P., Roubaud, D., 2017a. Bitcoin for energy commodities before and after the December 2013 crash: diversifier, hedge or safe haven. Appl. Econ. 49, 5063–5073. https://doi.org/10.1080/00036846.2017.1299102.
Bouri, E., Molnár, P., Azzi, G., Roubaud, D., Hagfors, L.I., 2017b. On the hedge and safe haven properties of bitcoin: is it really more than a diversifier? Finance Res. Lett. 20, 192–198. https://doi.org/10.1016/j.frl.2016.09.025.
Bouri, E., Lucey, B., Roubaud, D., 2019. Cryptocurrencies and the downside risk in equity investments. Finance Res. Lett. https://doi.org/10.1016/j.frl.2019.06.009. (in press).

Caporin, M., Costola, M., 2019. Asymmetry and leverage in GARCH models: a news impact curve perspective. Appl. Econ. 51, 3345–3364. https://doi.org/10.1080/00036846.2019.1578853.
Cheah, E., Fry, J., 2015. Speculative bubbles in Bitcoin markets? An empirical investigation into the fundamental value of Bitcoin. Econ. Lett. 130, 32–36. https://doi.org/10.1016/j.econlet.2015.02.029.
Chohan, U.W., 2019. Are Stable Coins Stable? Notes on the 21st Century (CBRi). https://doi.org/10.2139/ssrn.3326823.
Ciner, C., Gurdgiev, C., Lucey, B.M., 2013. Hedges and safe havens: an examination of stocks, bonds, gold, oil and exchange rates. Int. Rev. Finan. Anal. 29, 202–211. https://doi.org/10.1016/j.irfa.2012.12.001.
Corbet, S., Lucey, B., Urquhart, A., Yarovaya, L., 2019. Cryptocurrencies as a financial asset: a systematic analysis. Int. Rev. Finan. Anal. 62, 182–199. https://doi.org/10.1016/j.irfa.2018.09.003.
Corbet, S., Lucey, B., Yarovaya, L., 2018. Datestamping the Bitcoin and Ethereum bubbles. Finance Res. Lett. 26, 81–88. https://doi.org/10.1016/j.frl.2017.12.006.
Dar, A.B., Maitra, D., 2017. Is gold a weak or strong hedge and safe haven against stocks: robust evidences from three major gold-consuming countries. Appl. Econ. 49, 5491–5503. https://doi.org/10.1080/00036846.2017.1310998.
Dastgir, S., Demir, E., Downing, G., Gozgor, G., Lau, C.K.M., 2019. The causal relationship between bitcoin attention and bitcoin returns: evidence from the copula based granger causality test. Finance Res. Lett. 28, 160–164. https://doi.org/10.1016/j.frl.2018.04.019.
Davison, A.C., Smith, R.L., 1990. Models for exceedances over high thresholds. J. R. Stat. Soc.: Ser. B Methodol. 52, 393–425. https://doi.org/10.1111/j.2517-6161.1990.tb01796.x.
Engle, R., 2002. Dynamic conditional correlation: a simple class of multivariate generalized autoregressive conditional heteroskedasticity models. J. Business Econ. Stat. 20, 339–350. https://doi.org/10.1198/073500102288618487.
Feng, W., Wang, Y., Zhang, Z., 2018. Can cryptocurrencies be a safe haven: a tail risk perspective analysis. Appl. Econ. 50, 4745–4762. https://doi.org/10.1080/00036846.2018.1466993.
Fry, J., Cheah, E., 2016. Negative bubbles and shocks in cryptocurrency markets. Int. Rev. Finan. Anal. 47, 343–352. https://doi.org/10.1016/j.irfa.2016.02.008.
Gürgün, G., Ünalmış, I., 2014. Is gold a safe haven against equity market investment in emerging and developing countries? Finance Res. Lett. 11, 341–348. https://doi.org/10.1016/j.frl.2014.07.003.
Katsiampa, P., 2019. An empirical investigation of volatility dynamics in the cryptocurrency market. Res. Int. Business Finance 50, 322–335. https://doi.org/10.1016/j.ribaf.2019.06.004.
Kliber, A., Marszalek, P., Musialkowska, I., Świerczyńska, K., 2019. Bitcoin: safe haven, hedge or diversifier? Perception of bitcoin in the context of a country's economic situation – A stochastic volatility approach. Physica A: Stat. Mech. Appl. 524, 246–257. https://doi.org/10.1016/j.physa.2019.04.145.
Kroner, K.F., Ng, V.K., 1998. Modeling asymmetric movements of asset returns. Rev. Finan. Stud. 11, 844–871. https://doi.org/10.1093/rfs/11.4.817.
McAleer, M., 2014. Asymmetry and leverage in conditional volatility models. Econometrics 2, 145–150. https://doi.org/10.3390/econometrics2030145.
Micky News, 2019. Why New Stablecoins Are Being Developed at a Rapid Rate. (Accessed 5 February 2020). https://micky.com.au/why-new-stablecoins-are-being-developed-at-a-rapid-rate/.
Mita, M., Ito, K., Ohsawa, S., Tanaka, H., 2019. What is Stablecoin? A Survey on Price Stabilization Mechanisms for Decentralized Payment Systems. arXiv:1906.06037.
Omane-Adjepong, M., Alagidede, I.P., 2019. Multiresolution analysis and spillovers of major cryptocurrency markets. Res. Int. Business Finance 49, 191–206. https://doi.org/10.1016/j.ribaf.2019.03.003.
Patton, A.J., 2006. Modelling asymmetric exchange rate dependence. Int. Econ. Rev. 47, 527–556. https://doi.org/10.1111/j. 1468-2354.2006.00387.x.
Ranaldo, A., Söderlind, P., 2010. Safe haven currencies. Rev. Finance 14, 385–407. https://doi.org/10.1093/rof/rfq007.
Ratner, M., Chiu, C.-C.J., 2013. Hedging stock sector risk with credit default swaps. Int. Rev. Finan. Anal. 30, 18–25. https://doi.org/10.1016/j.irfa.2013.05.001.
Reboredo, J.C., 2013a. Is gold a hedge or safe haven against oil price movements? Resour. Policy 38, 130–137. https://doi.org/10.1016/j.resourpol.2013.02.003.
Reboredo, J.C., 2013b. Is gold a safe haven or a hedge for the US dollar? Implications for risk management. J. Banking Finance 37, 2665–2676. https://doi.org/10.1016/j.jbankfin.2013.03.020.
Selmi, R., Mensi, W., Hammoudeh, S., Bouoiyour, J., 2018. Is Bitcoin a hedge, a safe haven or a diversifier for oil price movements? A comparison with gold. Energy Econ. 74, 787–801. https://doi.org/10.1016/j.eneco.2018.07.007.
Shahzad, S.J.H., Bouri, E., Roubaud, D., Kristoufek, L., Lucey, B., 2019. Is bitcoin a better safe haven investment than gold and commodities? Int. Rev. Finan. Anal. 63, 322–330. https://doi.org/10.1016/j.irfa.2019.01.002.
Shen, D., Urquhart, A., Wang, P., 2019. Does twitter predict Bitcoin? Econ. Lett. 174, 118–122. https://doi.org/10.1016/j.econlet.2018.11.007.
Sidorenko, E.L., 2020. Stablecoin as a new financial instrument. In: Ashmarina, S., Vochozka, M., Mantulenko, V. (Eds.), Digital Age: Chances, Challenges and Future. ISCDTE 2019. Lecture Notes in Networks and Systems, vol. 84. Springer, Cham, pp. 630–638. https://doi.org/10.1007/978-3-030-27015-5_75.
Sifat, I.M., Mohamad, A., Mohamed Shariff, M.S.B., 2019. Lead-Lag relationship between Bitcoin and Ethereum: Evidence from hourly and daily data. Res. Int. Business Finance 50, 306–321. https://doi.org/10.1016/j.ribaf.2019.06.012.
Smales, L.A., 2019. Bitcoin as a safe haven: is it even worth considering? Finance Res. Lett. 30, 385–393. https://doi.org/10.1016/j.frl.2018.11.002.
Śmiech, S., Papież, M., 2017. In search of hedges and safe havens: revisiting the relations between gold and oil in the rolling regression framework. Finance Res. Lett. 20, 238–244. https://doi.org/10.1016/j.frl.2016.10.006.
Stensás, A., Nygaard, M.F., Kyaw, K., Treepongkaruna, S., 2019. Can Bitcoin be a diversifier, hedge or safe haven tool? Cogent Econ. Finance 7, 1593072. https://doi.org/10.1080/23322039.2019.1593072.
Urquhart, A., Zhang, H., 2019. Is bitcoin a hedge or safe haven for currencies? An intraday analysis. Int. Rev. Finan. Anal. 63, 49–57. https://doi.org/10.1016/j.irfa.2019.02.009.
Vidal-Tomás, D., Iba¨nez, A., 2018. Semi-strong efficiency of Bitcoin. Finance Res. Lett. 27, 259–265. https://doi.org/10.1016/j.frl.2018.03.013.
Wang, G.-J., Xie, C., Wen, D., Zhao, L., 2019a. When Bitcoin meets economic policy uncertainty (EPU): measuring risk spillover effect from EPU to Bitcoin. Finance Res. Lett. 31, 489–497. https://doi.org/10.1016/j.frl.2018.12.028.
Wang, G.-J., Tang, Y., Xie, C., Chen, S., 2019b. Is bitcoin a safe haven or a hedging asset? Evidence from China. J. Manag. Sci. Eng. 4, 173–188. https://doi.org/10.1016/j.jmse.2019.09.001.
Wang, P., Zhang, W., Li, X., Shen, D., 2019c. Is cryptocurrency a hedge or a safe haven for international indices? A comprehensive and dynamic perspective. Finance Res. Lett. 31, 1–18. https://doi.org/10.1016/j.frl.2019.04.031.
Wen, X., Cheng, H., 2018. Which is the safe haven for emerging stock markets, gold or the US dollar? Emerg. Markets Rev. 35, 69–90. https://doi.org/10.1016/j.ememar.2017.12.006.
Yi, S., Xu, Z., Wang, G.-J., 2018. Volatility connectedness in the cryptocurrency market: is Bitcoin a dominant cryptocurrency? Int. Rev. Finan. Anal. 60, 98–114. https://doi.org/10.1016/j.irfa.2018.08.012.
Zargar, F.N., Kumar, D., 2019. Informational inefficiency of Bitcoin: a study based on high-frequency data. Res. Int. Business Finance 47, 344–353. https://doi.org/10.1016/j.ribaf.2018.08.008.
Zhang, W., Wang, P., Li, X., Shen, D., 2018. The inefficiency of cryptocurrency and its cross-correlation with Dow Jones Industrial Average. Physica A: Stat. Mech. Appl. 510, 658–670. https://doi.org/10.1016/j.physa.2018.07.032.